

Project Blueprint: AgriPulse Telemetry Platform

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1. Project Overview

AgriPulse is an end-to-end, containerized IoT telemetry and supply chain monitoring platform. Designed to bridge the gap between real-world hardware data and enterprise cloud dashboards, this project monitors perishable goods (e.g., medical vaccines or fresh produce) in transit. It tracks critical metrics like temperature, humidity, and location to provide real-time alerts and historical analytics.

Goal: Demonstrate proficiency in cloud-native architecture, event-driven systems, and data pipelines to excel in the competitive Melbourne software engineering job market.

2. Technical Architecture

- **Ingestion:** Python-based IoT sensor simulation publishing telemetry via MQTT (Eclipse Mosquitto broker).
- **Backend:** **FastAPI**. An asynchronous, high-performance API for telemetry ingestion, threshold monitoring, and data serving.
- **Database:** **PostgreSQL**. Relational storage for long-term telemetry records and event logs.
- **Frontend:** **Next.js (React) + Tailwind CSS**. A dynamic, responsive dashboard for real-time visualization and alert management.
- **Infrastructure:** Containerized via **Docker Compose** for local development and simplified cloud deployment.

3. Key Engineering Focus

This project emphasizes production-grade standards:

- **Event-Driven Design:** Using MQTT for lightweight, real-time message broadcasting.

- **Asynchronous Processing:** Utilizing FastAPI's async capabilities to handle stream ingestion without blocking.
- **DevOps Best Practices:** Complete environment reproducibility via Docker.